

The Focusing Cubic NLS Soliton: Exact Hessian Spectrum and Pöschl–Teller Factorization

Route-A structural certificate HCS-C221

Abstract

We close a convention-locked one-dimensional theorem for the focusing cubic nonlinear Schrödinger equation. The sech standing wave has exact mass, Hamiltonian, action and Vakhitov–Kolokolov (VK) slope. Its two real Hessians have the complete Pöschl–Teller discrete spectrum, essential threshold, kernels and Morse index. The zero-frequency and defocusing faces are separated, and no finite-box spectrum or arithmetic interpretation is used. The variational scaling and explicit eigenfunctions are included.

Keywords: nonlinear Schrödinger; soliton; Hessian; Pöschl–Teller; Morse index; Route A.

1 Equation and standing wave

We study

$$i\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0, \quad \psi(t, x) = e^{i\omega t}Q_\omega(x), \quad \omega > 0, \quad (1)$$

on $\mathbb{R}$. Direct differentiation of $Q_\omega(x) = \sqrt{\omega} \operatorname{sech}(\sqrt{\omega}x)$ gives

$$-Q_\omega'' + \omega Q_\omega - 2Q_\omega^3 = 0. \quad (2)$$

For

$$M(u) = \int_{\mathbb{R}} |u|^2 dx, \quad H(u) = \frac{1}{2} \int_{\mathbb{R}} |u_x|^2 dx - \frac{1}{2} \int_{\mathbb{R}} |u|^4 dx, \quad S_\omega = H + \frac{\omega}{2} M,$$

the substitution $y = \sqrt{\omega}x$, $z = \tanh y$ yields

$$M(Q_\omega) = 2\sqrt{\omega}, \quad \|Q_\omega'\|_2^2 = \frac{2}{3}\omega^{3/2}, \quad \|Q_\omega\|_4^4 = \frac{4}{3}\omega^{3/2}, \quad (3)$$

and hence

$$H(Q_\omega) = -\frac{1}{3}\omega^{3/2}, \quad S_\omega(Q_\omega) = \frac{2}{3}\omega^{3/2}, \quad \frac{d}{d\omega}M(Q_\omega) = \omega^{-1/2} > 0. \quad (4)$$

2 The complete Hessian spectrum

The real and imaginary second variations are the self-adjoint operators

$$L_+ = -\partial_x^2 + \omega - 6\omega \operatorname{sech}^2(\sqrt{\omega}x), \quad L_- = -\partial_x^2 + \omega - 2\omega \operatorname{sech}^2(\sqrt{\omega}x). \quad (5)$$

The theorem is

$$\sigma_{\text{ess}}(L_+) = \sigma_{\text{ess}}(L_-) = [\omega, \infty), \quad \sigma_{\text{disc}}(L_+) = \{-3\omega, 0\}, \quad \sigma_{\text{disc}}(L_-) = \{0\}. \quad (6)$$

The -3ω eigenfunction is $\operatorname{sech}^2(\sqrt{\omega}x)$; the zero eigenfunctions are Q_ω' for L_+ and Q_ω for L_- . All listed eigenvalues are simple. Thus L_+ has Morse index one and kernel $\operatorname{span}\{Q_\omega'\}$, while $L_- \geq 0$ and has kernel $\operatorname{span}\{Q_\omega\}$.

To see the signs without a box discretization, put $y = \sqrt{\omega}x$ and

$$A_\ell = \partial_y + \ell \tanh y, \quad A_\ell^* = -\partial_y + \ell \tanh y.$$

Multiplication gives

$$-\partial_y^2 + 1 - 6 \operatorname{sech}^2 y = A_2^* A_2 - 3, \quad -\partial_y^2 + 1 - 2 \operatorname{sech}^2 y = A_1^* A_1. \quad (7)$$

The $\ell = 1, 2$ ladder has exactly the bound states displayed in (6); the free remainder starts at one, which rescales to the threshold ω .

3 Boundaries and evidence

At $\omega \downarrow 0$, the profile and discrete eigenvalues collapse to the essential threshold. Reversing the cubic sign has no nonzero bright H^1 sech branch. A periodic domain gives elliptic/cnoidal waves, and dimensions $d \geq 2$ have different criticality; neither is silently included.

Revision focus. Round 1 adds (3)–(6), explicit kernels and the zero-frequency/defocusing boundary.

References

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Declarations

Data and code. Synthetic exact probes and package-local deterministic code accompany the manuscript; no training data are used. **Ethics/funding.** No human or animal data; no external funding reported. **AI disclosure.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.